

Hol. XX

No. 2

The Fern Bulletin

A Quarterly Devoted to Ferns



Joliet, Ill.

Willard N. Clute & Company

1912

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BULLETIN, APRIL 1912 ***

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April

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GYMNOGRAMMA LANCEOLATA

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THE FERN BULLETIN

THE FERN FLORA OF ILLINOIS.

BY E. J. HILL.

The state of Illinois has an area of about 55,000 square miles. It lies between the parallels 37° and $42^{\circ} 30'$, thus giving a length of $5\frac{1}{2}^{\circ}$ or about 380 miles. This north and south extension produces a milder climate in the southern part, but no fern of essentially southern distribution comes in except *Polypodium polypodioides*, though the two quill-worts of the state are perhaps better placed under this head also. It is the lowest of the north-central states in average altitude, the mean above sea level being about 600 feet, varying from 300 feet at the junction of the Ohio and Mississippi rivers to 1250 feet at the Wisconsin line in the extreme northwest part. As there is nothing in these extremes of elevation to effect material changes of temperature due to altitude, its floristic features are not much modified in respect of this. Anything of this character must be ascribed to local conditions, not general causes. Another factor that affects its floristic features is the dominance of prairie within its boundary, the forests and woodlands, sometimes very narrow strips, chiefly bordering its streams and lakes. Since lands covered with grass are not adapted to the growth of ferns, and consequently are limited in species, their number and variety must be much restricted for this reason. This must have been the case in the primitive condition of the prairies before they were so generally taken up for cultivation. The loss in the original fern-flora is slight in this regard when compared with that of flowering plants. As nearly all of the state is in the region of the glacial drift, the soil is influenced by this condition also. The ravines cut in the drift and in the underlying rock where it is reached, with their varying degrees of moisture and shade, show the greatest variety in fern-life, though a greater abundance of certain kinds may be found in woods and swamps. The prevailing rocks are limestone, but sandstones occur in some localities, especially along the Illinois and Rock rivers. These in some parts of the state, particularly in the coal measures, the area of which is large, may be interstratified with shales and slate. These rocks and the soils resulting from their disintegration and decomposition, taken in connection with those of the glacial drift, provide a

fair range of edaphic conditions for the growth of ferns. It is evident that such as prefer a calcareous soil will be best represented, if any preference of this kind inheres in their nature.

It will be seen from the list that not quite one half (56) of the Pteridophytes accorded specific rank in "Gray's New Manual of Botany" (115) are reported from this state. The genera are represented in larger proportion, 23 of the 31 given, or if *Athyrium* be separated, 24 of 32, or three-fourths of them. All the species of several of the smaller genera are found, up to three in the case of *Osmunda*, but all of none with species exceeding this number. The genus most fully represented is *Equisetum*, eight of the ten, or nine of eleven when *E. robustum* is given specific rank. To these must be added *E. Ferrissii*, not in the Manual.

Reliable data for the distribution of the ferns of the state are not very full. It is hoped that they may be made more complete by the co-operation of those into whose hands the list may fall. Many additions to the number of 34 species can hardly be expected. Doubtless the state has been quite well explored in this respect. I find only two to add to those published by Patterson in 1876, *Isoetes Butleri*, described in 1878 from specimens found in Indian Territory (Oklahoma) but since found in this state, and *Equisetum Ferrissii*, a recent addition. The list is mainly a compilation made at the request of the editor of the *Fern Bulletin*. No special fitness for the task is claimed, since my personal knowledge of the region covered is almost wholly confined to five of the northeastern counties, Kankakee, Will, Cook, Dupage and Lake. Only casual trips of slight duration have been made to other places. The publication most relied on for the state at large is the "Catalogue of the Phaenogamous and vascular cryptogamous plants of Illinois," H. N. Patterson, Oquauka, Ill., 1876. His catalogue of plants growing in the immediate vicinity of Oquauka has also been used. Friedrich Brendel's "Flora Peoriana, Budapest, 1882," (the German edition, but since given in English, I believe) has furnished some definite information for a district around the city of Peoria. The floras of H. H. Babcock and of Higley and Raddin for Chicago and vicinity have likewise been consulted, but as they respect territory mainly familiar to the writer, could be cited but little.

As explanatory of the plan followed I may state that I have first mentioned the localities or stations with which I am personally acquainted, and from which examples are in my herbarium unless very common throughout. Citations from Patterson's catalogue for the state at large are entered in quotation marks followed by (P.). Where Peoria is given the authority is Brendel, where Oquauka, Patterson. A few have been furnished by V. 35 H. Chase, who collected in Stark county and vicinity, and by Prof. Atwell of the Northwestern University, from data in the herbarium of the University.

OPHIOGLOSSACEAE.

OPHIOGLOSSUM VULGATUM (L.) "Wabash county, a single plant." *Schneck*. (P.) Probably elsewhere, but easily overlooked.

BOTRYCHIUM OBLIQUUM (Muhl.) In open woods, Cook Co., rare. "S. Illinois. *Vasey, Schneck*." (P.) Peoria Co., *V. H. Chase*. Starved Rock. *J. H. Ferriss*.

BOTRYCHIUM OBLIQUUM DISSECTUM. (Spreng.) Peoria Co., *V. H. Chase*.

BOTRYCHIUM VIRGINIANUM. (L.) Common in rich woods in the northeastern part of the state, and probably throughout. It often occurs in colonies, sometimes of a dozen or more plants. In woods along Lake Michigan it reaches a height of two feet.

OSMUNDACEAE.

OSMUNDA CINNAMOMEA. (L.) Abundant in swampy areas in the northeastern counties, especially in peaty ground near Lake Michigan within the limits of the ancient glacial Lake Chicago. Swampy areas in sand barrens west of Kankakee, "Menard county. *Hall*." (P.) Starved Rock. *Clute*.

OSMUNDA CLAYTONIANA (L.) Frequent in swamps and wet woods from Kankakee county north in the eastern part of the state. Peoria, *Brendel*.

Henderson Co., *Patterson*. "Moist ravines, common." says Patterson for the state at large.

OSMUNDA REGALIS (L.) Has a range similar to the last and is quite frequent northeast in swamps and wet woods. Peoria, *Brendel*. Mason county, *Bebb*. Infrequent says Patterson for the state as a whole. 36

POLYPODIACEAE.

ADIANTUM PEDATUM (L.) Common throughout the state in rich woods.

POLYPODIUM VULGARE (L.) On cliffs of sandstone, La Salle and Ogle counties. "Common in Jackson and Union, *French, Forbes*." (P.)

POLYPODIUM POLYPODIOIDES (L.) Common throughout the state in rich woods.

PTERIS AQUILINA (L.) Copses and borders of dry woods. Frequent, or abundant in localities northeast. Starved Rock, La Salle county, Peoria, *Brendel*, Henderson, *Patterson*, Shelby, *Mary Evertz*. "Common." for the state. (P.) Rare in Will county in the prairie region. *Clute*.

CHEILANTHES LANOSA (Michx.) "Rocks, St. Clair county, *Brendel*, and southward." (P.)

CHEILANTHES FEEI (Moore.) Limestone cliffs by Mississippi river, Carroll county, "near Galena, *Brendel*; Pike county, *Mead*; Jackson, *French*." (P.)

PELLAEA ATROPURPUREA (L.) Frequent on cliffs of limestone along the Desplaines river and its tributaries from Sag Bridge, Cook county, to Joliet, Will county, and in Kankakee and Carroll counties. Scarce on cliffs of sandstone, Oregon, Ogle county. Henderson county, *Patterson*; Kane county, *W. J. Minium*; Wedron, La Salle county, *Ferriss*. Reported for the state as general but "infrequent" in Patterson's catalogue.

PELLAEA GRACILIS (Michx.) Rare in thin soil in shelves of shaded and usually moist calcareous rocks. Sag Bridge and Lemont, Cook county, and Bounbonnais, Kankakee county. On moist sandstone rocks, Liberty Hill, Oregon, Ogle county; limestone, Aurora, Kane county; sandstone, Sheridan, La Salle county, *Ferriss*. 37

ASPLENIUM ANGUSTIFOLIUM (Michx.) Henderson. *Patterson*, Peoria, *Brendel* "Rich woods, scarce for the state." (P.) Joliet rare, Starved Rock more common. *Ferriss*.

ASPLENIUM PINNATIFIDUM (Nutt.) "On rocks, Jackson and Union counties, *French*; Pope, *Schneck*." (P.)

ASPLENIUM PLATYNEURON (L.) "Open rocky woods, scarce." (P.)

ASPLENIUM EBENOIDES (R. R. Scott.) Reported from Jackson county, Ill., but without further reference in *Fern Bulletin*, vol. V., p. 13.

ASPLENIUM TRICHOMANES (L.) "On shaded rocks, Jackson and Union counties, *French*; Wabash, *Schneck*." (P.) Southern Illinois. *Vasey*. Starved Rock, two plants. *Ferriss*.

ATHYRIUM FILIX-FOEMINA (L.) Frequent in rich, moist woods in Cook and adjoining counties, as well as throughout the state as given by *Patterson*, Peoria, *Brendel*; Jackson, *Saml. Bartley*; Henderson, *Patterson*; Ravinia, Willow Springs, Cook county, *Prince*.

ATHYRIUM THELYPTEROIDES (Michx.) "Near Glencoe, Cook county." *Higley Raddin*; "Peoria and Fulton counties, *Brendel* and *Wolff*; Wabash, *Schneck*." (P.) Joliet, rare; Starved Rock abundant, *Ferriss*.

CAMPTOSORUS RHYZOPHYLLUS (L.) On outcrops of limestone in the Desplaines valley in Cook and Will counties from Sag Bridge to Joliet. Abundant at Dellwood Park and in one locality at Sag Bridge, infrequent elsewhere. "Shaded rocks throughout but scarce." (P.) Jo Daviess county, *Pepoon*. 38

PHEGopteris hexagonoptera (Michx.) “Rich open woods and shaded ravines, chiefly in the northern portion of Cook county; infrequent.” *Higley* and *Raddin* (1891.) Peoria, *Brendel*; Henderson, *Patterson*; Jackson, *Bartley*; Joliet and Starved Rock, *Ferriss*. Patterson reports “frequent” throughout.

PHEGopteris polypodioides (Fée.) Starved Rock, La Salle county, “Menard county, *Hall*.” (P.)

Nephrodium noveboracense (L.) “Elgin, Kane county, *Vasey*; Wabash, *Schneck*, Swamps, scarce.” (P.)

Nephrodium thelypteris (L.) Frequent or often abundant in swampy, wooded ground or open marshes, in Cook, Lake, Dupage, Will and Kankakee counties, Peoria, *Brandel*; Starved Rock, *Clute*. Frequent throughout the state according to Patterson.

Nephrodium cristatum (Michx.) Starved Rock, rare, *Ferriss*.

Nephrodium goldieanum (Hook.) “Rich Woods, Peoria and Fulton counties, *Brendel*, *Wolff*; Makanda, Jackson county, *Forbes*,” (P.) Will county, La Salle county, *Ferriss*.

Nephrodium marginale (L.) Rocky bluffs, Starved Rock, La Salle county, Southern Illinois, *Vasey*. “Scarce” for the state. (P.)

Nephrodium spinulosum intermedium (Muhl.) Frequent in rich woods in the northeastern counties, Starved Rock, *Clute*. Patterson says “infrequent” for the state.

Polystichum acrostichoides (Michx.) Will county, “north part of Cook county,” *Higley* and *Raddin*; Henderson, *Patterson*; Peoria, *Brendel*; Jackson, *Bartley*. For the state, “infrequent.” (P.) The variety *incisum* is occasionally reported.

Cystopteris bulbifera (L.) Frequent on shelves and in crevices of limestone cliffs and shady ravines in the Desplaines valley in Cook and Will counties, and in Kankakee county, Henderson, *Patterson*, Peoria,

Brendel; Starved Rock, abundant, *Clute*. Patterson reports for the state, “shaded rocks, frequent.”

CYSTOPTERIS FRAGILIS (L.) Rather frequent in rich woods and occasionally on rocks in Cook, Lake, Dupage, Will and Kankakee counties; Henderson, *Patterson*; Peoria, *Brendel*; Jackson, *Bartley*. “Common” for the state. (P.) Very variable in its forms.

WOODSIA OBTUSA (Spreng.) Scarce on limestone rocks at Lemont, Cook county, abundant on sandstone at Oregon, Ogle county, “Marion county, *Bebb*; Wabash, *Schneck*; and southward.” (P.) Joliet, Will county, *Ferriss*.

WOODSIA ILVENSIS (L.) “On sandstone cliffs near Oregon, Ogle county, *Bebb*.” (P.)

ONOCLEA SENSIBILIS (L.) Common in wet woods and swamps in the northeastern counties. Peoria, *Brendel*; Jackson, *Bartley*. For the state “common.” (P.)

ONOCLEA STRUTHIOPTERIS (L.) Wet shades, Starved Rock, La Salle county, Henderson, *Patterson*; Peoria, *Brendel*; Fulton, *Wolff*. For the state “infrequent.” (P.)

DICKSONIA PUNCTILOBULA (Michx.) “Wabash county, *Schneck*.” (P.)

SALVINIACEAE.

AZOLLA CAROLINIANA (Willd.) “Ponds from Henderson and Peoria counties southward. Infrequent.” (P.) “Since 1857 not found again in the region of our local flora.” *Brendel* in Flora Peoriana. “In a pond near South Chicago, 1886. So far as known this is the only locality where this species has been found within our limits.” *Higley* and *Raddin*.

EQUISETACEAE.

EQUISETUM ARVENSE (L.) Common from Kankakee county north. Reported by Patterson as common throughout the state. Though usually growing in moist sand or gravel, it is often found in the Chicago region in masses along dry railway embankments.

EQUISETUM PALUSTRE (L.) “Wet places. Peoria county, *Wolff, Brendel*.” (P.)

EQUISETUM FLUVIATILE (L.) In shallow water or very wet ground. Quite frequent about Chicago. “Cass county, *Mead*; Peoria, *Brendel*; McHenry, *Vasey*. Scarce.” (P.) Joliet, common, *Ferriss*.

EQUISETUM LAEVIGATUM (A. Br.) Cook and Kankakee counties. “In dry or moist clay or sand from Henderson and Peoria counties southward.” (P.) In the Chicago region generally in moist sands; Hancock county, *Mead*.

EQUISETUM HYEMALE (L.) Moist places. Cook, Will and Lake counties. Frequent, as well as throughout the state according to Patterson.

EQUISETUM FERRISSII (Clute.) Moist banks, Will county.

EQUISETUM ROBUSTUM (A. Br.) On moist or wet banks of streams. Thornton and La Grange, Cook county. “River banks from Peoria county southward.” (P.)

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EQUISETUM VARIEGATUM (Schleich.) In clayey ravines at Lake Forest and in wet sands at Waukegan, Lake county, Peoria, *Brendel*. Var. *Jesupi*, A. A. Eaton, and var. *Nelsoni*, A. A. Eaton, are credited to Illinois in Gray’s New Manual of Botany. The latter variety occurs in Lake county, Ind., bordering Illinois, and is likely to be found in the neighboring parts of this state, but those from Lake county, Ill., agree better with the typical form.

EQUISETUM SCIRPOIDES (Michx.) Moist shaded ravines, Lake Bluff, Lake county. Reported by *Cowles* at Lake Forest. “Ringwood, McHenry county, *Vasey*.” (P.)

LYCOPODIACEAE.

LYCOPODIUM INUNDATUM (L.) "Moist sands, south Evanston, Cook county."
Higley and Raddin.

LYCOPODIUM LUCIDULUM (Michx.) "Moist woods, Evanston, Cook county,
Vasey; Ogle, Bebb." (P.)

LYCOPODIUM SELAGO (L.) "Collected by J. W. Powell near Ottawa, *Vasey.*"
(P.)

SALAGINELLACEAE.

SELAGINELLA RUPESTRIS (L.) Dry sands and sandstone rocks, La Salle and
Ogle counties. "Dry rocks and barrens, Henderson county; Ogle. *Bebb,*
Rare, or overlooked." (P.)

SELAGINELLA APUS (L.) Low sandy, peaty, or springy ground, Kankakee,
Cook, Lake and Will counties. Peoria, *Brandel*; Lawns in Joliet, *Miss*
L. M. Hird. "Low sandy places," says Patterson, as if throughout the
state.

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ISOETACEAE.

ISOETES MELANOPODA (J. Gay.) "Muddy borders of a pond near Hyde Park
water-works, 1885. Wet prairies near Grand Crossing, 1886-87."
Higley and Raddin. These stations in Cook county are doubtless
destroyed now. Stark county, *V. H. Chase.* "Menard, *Hall*; Fulton,
Wolff; McHenry, *Vasey.*" (P.)

ISOETES BUTLERI (Engelm.) "Moist hillsides and shallow depressions, Illinois
and Kansas to Tennessee and Oklahoma." Gray's New Manual of
Botany.

A PROBLEMATICAL FERN. (*Gymnogramma lanceolata*.)

BY WILLARD N. CLUTE.

In the identification of fern species one occasionally comes upon two forms so nearly alike that it requires very careful study to decide whether they are two different species or merely two forms of a single variable species, but it is rare that one finds a fern that can as well be placed in one genus as another, and still more rare when the species possesses characters so like those of ferns in other groups that it may be moved from one tribe to another without violating any of the botanical properties. The fern chosen for illustration here is one of this latter character. It has been passed back and forth between various genera in different tribes, seldom resting long in one place, until it is a very problematical species indeed.

In outline and manner of growth it possesses no especial peculiarities. The lanceolate leaves might fit any of a dozen or more species that might be mistaken for it if the fruit dots or sori were absent. *Vittaria*, *Taemitis*, 43 *Antrophyum*, *Polypodium*, *Asplenium*, *Acrostichum* and many other genera have species with leaf outlines that almost exactly match it, but a glance at the fruiting fronds, at once excludes many of these genera as possible harbors for the species and at the same time increases the difficulties of finally placing it. The sori are apparently linear and *Scolopendrium* or *Asplenium* comes to mind, but there is no indusium and so the relationship is thrown into that group of ferns clustering about such forms as *Gymnogramma*.

In fact, our fern was for a long time known as *Gymnogramma lanceolata* and owing to this fact I have selected this to stand as the name of the plant. A glance at the illustration, however, will disclose a frond not at all like the conventional *Gymnogramma* frond, but it is as much like a *Gymnogramma* as it is like the family to which the plant is now assigned. Curious as it may seem this plant with elongated sori oblique to the midrib is now regarded as a *Polypodium*! Before its settling down in this genus, it had been placed in

Antrophyum, *Grammitis*, *Loxogramme* and *Selliguea* as well as *Gymnogramma*. This is by no means due to the variable nature of the fern. Through all these vicissitudes it has remained unchanged. The fluctuations from one genus to another even from one tribe to a different one, have been due to the varying opinions of mere man and his efforts to fit the fern to a set of descriptions of his own making. Circumstances such as these are quite sufficient to justify the refusal to accept off-hand the results of every “revision” which ambitious systematists see fit to inflict upon us.

While reposing in the genus *Gymnogramma*, the fern was well-known to be somewhat unorthodox. In every large assemblage of species there are, in addition to those which are typical, certain others that diverge 44 somewhat, but not enough to form a separate genus. Thus our plant was placed in the section *Selliguea*. Sometimes, indeed, *Selliguea* was isolated as a separate genus, but usually accompanied by the statement that if it were not for the shape of the sorus it would make a good addition to the section *Phymatodes* of *Polypodium*. Here, at least, is where it has landed, the elongated sori being winked at, possibly, or perhaps the species makers are willing to assume each so-called sorus to be a series of *Polypodium* sori. In this age, however, there are those who deny to the species in the group *Phymatodes* the right to be included in *Polypodium* and in certain books our species appears as *Phymatodes loxogramma*. Just how this *loxogramme* came to supplant *lanceolata* is another story, not to be detailed here. Suffice to say that the new name was picked up during one of the fern’s numerous transfers.

As to *Phymatodes*, it is likely that the species in this group are distinct enough to form a genus by themselves but it would be a rash student to encourage such a departure, for once started we should soon see all the large genera cut up into lesser groups and then what delightful times the name-tinker would have!

By what ever name called, the species manages to thrive over a wide stretch of country in the Eastern Hemisphere, being found from Japan and China to the Himalayas, Ceylon and the Guinea Coast and represented in many of the islands of the Pacific including Fiji and Samoa. The specimen from

which the illustration was made was collected by K. Miyake near Kyoto, Japan where it is reported “not so common.”

THE TALL SPLEENWORTS.

BY ADELLA PRESCOTT.

Some years ago when for me there were but two species of ferns, those that were finely cut and those that were not—and maidenhair—I supposed of course that the narrow leaved spleenwort (*Asplenium angustifolium*) was simply a hardy sword fern and that both were varieties of the Christmas fern! But when I began to read the fascinating pages of Clute and Parsons and Waters I found, even in the early summer, that there were differences and by the time the sori appeared I was wise enough to recognize the characteristic mark of the spleenworts. Even then I thought it but a common fern for in the woods with which I was most familiar it grew plentifully and it was not till sometime later that I learned that it is at least rare enough to insure for itself a welcome whenever found.

It is an extremely local plant and may be looked for perhaps for years before being found though it has a wide distribution and is apt to be plentiful where it grows at all. It prefers rather moist soil and seems to like Goldie's fern for a neighbor as I have often found them in close proximity.

The fronds grow in tufts from a creeping rootstock and are said to reach a height of four feet but all that I have seen were shorter by at least a foot. The blades are simply pinnate with many long, narrow pinnules tapering to slender tips. The fertile fronds are taller with the pinnules much narrower and the linear sori borne in two rows along the midrib of each pinnule. The fronds are delicate in texture and are easily destroyed by summer storms, yet the plant is able to adapt itself in some degree to its environment for a plant that I have in a border where it is exposed to cold winds has become much more rugged both in appearance and in fact. It is a charming addition to the fern garden making a pleasing foil to *Nephrodium spinulosum*, *Dicksonia* and other finely cut varieties.

I think it is a pity that the silvery spleenwort has no common name but one that is suggestive of a varied assortment of "blues," and that does not

certainly belong to it at that. But when we consider the discomforts suggested by the word “spleeny” we may think after all that this plain unassuming plant would prefer to be classed among the spleenworts with their fabled powers of healing rather than among the gentle folk of the *Athyriums* where perhaps it rightly belongs.

The silvery spleenwort, *Asplenium thelypteroides*, or *Athyrium thelypteroides* as some prefer to call it, has few characteristics that would make it noticeable among other species. It is of an ordinary size, from two to three feet in height, and the fronds are produced singly from a stout creeping rootstock but they grow so close together as to suggest a circular crown. They are once pinnate with deeply lobed pinnules and have rather a soft velvety texture though quite thin and delicate. The blade is oblong, tapering both ways from the middle and there is little difference between the fertile and sterile fronds.

The sori are borne in regular double rows on the pinnules and while in general they are like those of the spleenwort yet they are frequently curved after the fashion of the lady fern, making a puzzling question on which the botanical doctors fail to agree.

This species is fairly common over a wide area and while not possessing any striking beauty is interesting and attractive to the true lover of ferns.

New Hartford, N. Y.

FURTHER NOTES ON VARIATION IN BOTRYCHIUM RAMOSUM.

BY RAYNAL DODGE.

On June 2nd of the present year I again visited the *Botrychium* stations at Horse Hill, Kensington, N. H., and at Newfound Hill in Hampton Falls. A description of these was given in *The Fern Bulletin* April 1910. I found that a great change had taken place since my last visit in 1907. The young trees had grown wonderfully and shaded the station, the farm house had been abandoned, the hens had disappeared, and *Botrychium ramosum* had again taken its place at the foot of the hill. But instead of the many thousands which formerly grew there, I only succeeded in finding about forty plants, some of them however, quite robust and well grown. On the same day, in company with a friend, I made a thorough search for *Botrychium simplex* at Newfound Hill but failed to find a single plant.

It appears that all the forms in the genus *Botrychium* increase in numbers very slowly and that the individual plants require many years to attain their full development, but if the station for *Botrychium ramosum* on Horse Hill escapes damage by fire or marauding hens I think that within twenty years someone perhaps now younger than I, may find a large colony of *Botrychium simplex* at the old station on Newfound Hill. Several of my young friends have undertaken if possible to make a search.

Perhaps some of the readers of *The Fern Bulletin* know of localities where *Botrychium ramosum* and *B. simplex* are to be found growing near each other. If any such are known it seems that further investigations relating to this subject might be made. Or perhaps it would be enlightening if spores of *B. ramosum* in sufficient quantity were to be sown on some dry 48 hillside that was easily accessible to the experimenter. Immediate results however should not be expected as these *Botrychiums* move very slowly, according to some experimenters requiring several years before germination of the spores. Moreover in the present case the continued growth of the young plants would be very much dependent on the amount

of moisture they might receive as is evidenced by the total destruction of the plants at Newfound Hill by a very severe drouth.

Since speaking on this subject before the members of the American Fern Society I have been informed of two other instances besides those at that time mentioned where plants of *B. simplex* once found had disappeared which seems further evidence that the form *simplex* in *Botrychium* described by Hitchcock as growing in dry hills is not self-perpetuating.

Newburyport, Mass.

[To the instances of the disappearance of *B. simplex*, may now be added the disappearance of the colony found at Glen Park, Indiana in 1910. In that year there was perhaps a hundred plants found. Every year since, members of the Joliet Botanical Club and others have searched for them but not a single specimen has been discovered. Some *Botrychiums* have the habit of resting for a year or more, but it hardly seems likely that they would rest for three summers in succession.—*Ed.*]

RARE FORMS OF FERNWORTS—XXII.

STILL ANOTHER CHRISTMAS FERN.

In 1893, the late James A. Graves found a curious form of Christmas fern (*Polystichum acrostichoides*) in the vicinity of Susquehanna, Pa., and removed it to his garden where it continued to put forth its abnormal fronds for many years and may still be alive for anything the writer knows to the contrary. During the period in which Mr. Graves gave his principal attention to the study of ferns he was often advised to describe his abnormal specimen, but he was always so much engrossed in the study and cultivation of the living ferns that he never found time to write a formal scientific description of the plant, though he had settled on a name for it. The form undoubtedly deserves a distinctive name and since the discoverer is no longer with us, it seems very fitting that the form be named for him. I therefore offer the following description of 49

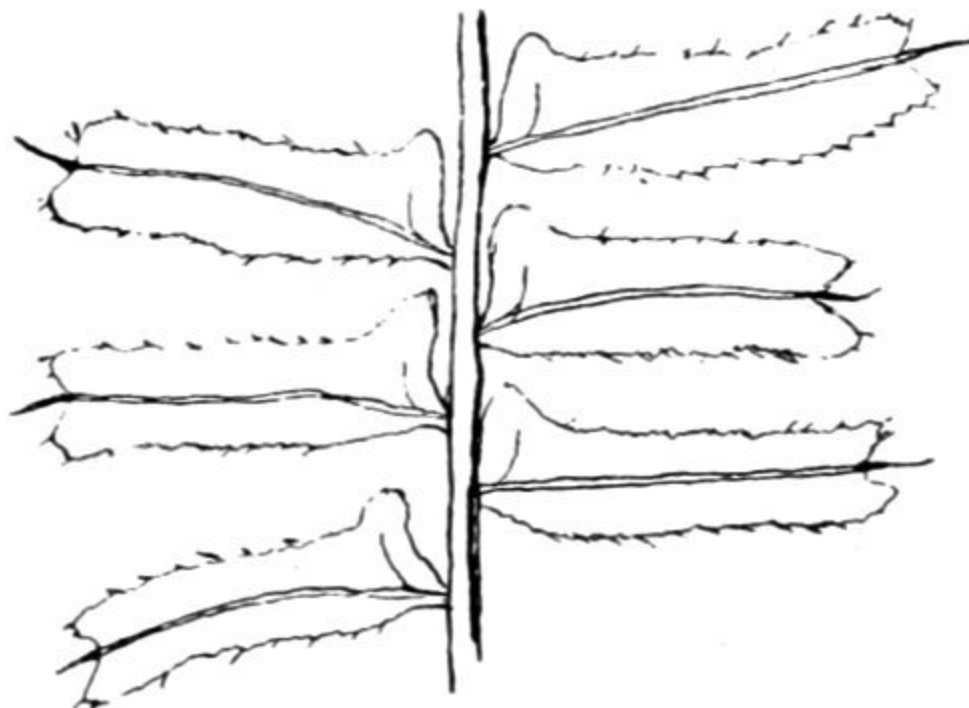
POLYSTICHUM ACROSTICHOIDES f. GRAVESII.

Plant similar to the type but with the pinnae ending in truncate tips from which the midveins project as spinelike bristles. Type in the herbarium of Willard N. Clute. Cotype in the herbarium of Alfred Twining, Scranton, Pa.

Although the description is drawn from a single plant it is likely that a search in the regions where the Christmas fern is abundant would reveal other specimens with the same peculiarity. Indeed, H. G. Rugg in a paper before the Vermont Botanical Club, last winter, described a plant that, to judge from his remarks must be essentially the same thing. He says: "For several years I have had a peculiar form of this fern growing in my garden. It is interesting because of the truncate form of the pinnae and the multifid form of the tip of the frond. The sterile fronds are usually like those of the type plant. This fern I transplanted into my garden several years ago and ever since then it has continued to bear these peculiar fronds. The late Mr.

B. D. Gilbert was interested in the plant and asked permission to describe it in the *Fern Bulletin* but illness and finally death prevented.” Apparently the only difference between the Vermont and Pennsylvania plants is the cristate apex, but as forking tips are to be expected in any species this feature is not extraordinary. 50

Mr. Graves usually spoke of his specimen as the variety *truncatum*. This is the name it bears in some herbaria and is the one it undoubtedly would have borne in literature had he lived to describe it. Those who were fortunate enough to have known Mr. Graves personally, however, will be pleased to see his name associated with one of the forms of that division of the plant world which he studied so long and so assiduously. It need hardly be said for the readers of this magazine that Mr. Graves was one of the founders of the Linnaean Fern Chapter the name by which the American Fern Society was originally known, was elected the first treasurer and held that office through half the lifetime of the society, was one time president of the same society and for a long time one of the most resourceful of its Advisory Council members.



The drawing herewith was made from the middle pinnae of a frond kindly supplied by Mr. Alfred Twining, of Scranton, Pa. It is a fair average of the form and though without much beauty of outline is still of interest for the form in which nature has cast it.

NOTES ON VARIOUS FERNS.

BY S. FRED PRINCE.

I was very much interested in Mr. Hill's article on the cliff brakes in the January Bulletin. I lived at Madison, Wisconsin, from 1874 to 1878, and have gathered *Pellaea atropurpurea* many times from the sandstone cliffs, not only on Lake Mendota, but also Lake Monona and outcrops in other parts of the "Four-lake County."

I found it growing on both the Potsdam and the Madison sandstones. On the former it was only in small clumps, or isolated plants, much more sparse in growth than when on the latter, though I never found it anywhere in such dense, tangled masses as it forms in the clefts of the limestone rocks of the southwest Ozarks.

I have also found *Pellaea atropurpurea* growing thinly, on a dark red sandstone, at Paris Springs, Missouri, not far from Springfield.

I would like to add to the localities of *Polypodium vulgare* in Michigan. I found it, in the summer of 1910, growing in dense mats on sand dunes, south of Macatawa, Michigan. The plants were in a woodland composed principally of hemlock, with oak and a general mixture of elm, maple, hickory, etc. When you lifted a mat of the fern, the bare sand was left exposed. I thought the conditions rather peculiar.

I found many ferns growing on these wooded sand hills where, at the most, there was but half an inch of soil on top of the white sand. The list includes:

Adiantum pedatum; *Pteris aquilina*; *Asplenium filix-foemina*, in 52 marshy places between the dunes; *Polystichum acrostichoides*, very sparingly; *Nephrodium thelypteris*, very luxuriant, like the lady fern, in marshy ground; *Nephrodium marginale*, the most common fern; *Nephrodium cristatum*; *Nephrodium spinulosum*, wherever there was a rotting chunk of wood; *Onoclea sensibilis*, and *Onoclea struthiopteris*, both

very rank; *Osmunda regalis* and *Osmunda cinnamomea*, these last four in marshy spots; and *Botrychium virginianum*, on the sides of the dunes.

I have been observing the habits of *Onoclea sensibilis* for many years, even raising plants from the spores to five years old; caring for other plants for years, changing conditions, and varying my experiments, until I have come to the following conclusions:

When the soil is constantly and evenly moist and unusually rich, and the plant is constantly shaded, it tends to produce its fertile fronds flattened out like the sterile, with all stages to those only partly rolled up. These *unrolled* fertile fronds do not differ from the *rolled up* ones, on the same plant, except in this one particular.

When a heavy screen was changed so that the plants would be in the full light and sun, the fertile fronds produced the rest of the season were as tightly rolled as usual, and it took two years of shading before these plants produced open or unrolled fertile fronds again. Varying the other conditions—moisture and nutriment, had similar results, but less marked.

Champaign, Ill.

SCHIZAEA PUSILLA AT HOME.

Anyone who has seen this odd fern growing in its native haunts will probably concur in the opinion held by some, that while it is looked upon as one of the rarest of ferns its small size and its habit of growing in the midst of other low plants have no doubt caused it to be passed over by collectors in many regions where it really exists. This should be an encouragement to collectors to keep the fern in mind in their field excursions with a view to adding new stations for it to those now known. The finding of a rare plant in a new locality is always a source of especial pleasure to the discoverer, aside from being an item of value to the botanist in general.

Schizaea pusilla was first collected early in this century at Quaker Bridge, N. J. about thirty-five miles east of Philadelphia. The spot is a desolate looking place in the wildest of the "pine barrens" where a branch of the Atsion river flows through marshy lowlands and cedar swamps. Here amid sedge grasses, mosses, *Lycopodiums*, *Droseras* and wild cranberry vines the little treasure has been collected. But though I have hunted for it more than once my eyes have never been sharp enough to detect its fronds in this locality.

In October of last year, however, a good friend guided me to another place in New Jersey where he knew it to be growing and there we found it. It was a small open spot in the pine barrens, low and damp. In the white sand grew patches of low grasses, mosses, *Lycopodium Carolinianum*, *L. inundatum* and *Pyxidanthera barbata*, besides several small ericaceous plants and some larger shrubs, such as scrub oaks, sumacs etc. Close by was a little stream and just beyond that a bog. Although we knew that *Schizaea* grew within a few feet of the path in which we stood, it required the closest 54 kind of a search, with eyes at the level of our knees before a specimen was detected. The sterile fronds, curled like corkscrews, grew in little tufts and were more readily visible than the fertile spikes which were less numerous and together with the slender stipes were of a brown color hardly distinguishable from the capsules of the mosses and the maturing stems of the grasses which grew all about. Lying flat upon the earth with face within

a few inches of the ground was found the most satisfactory plan of search. Down there all the individual plants looked bigger and a sidelong glance brought the fertile clusters more prominently into view. When the sight got accustomed to the miniature jungle, quite a number of specimens were found but the fern could hardly be said to be plentiful and all that we gathered were within a radius of a couple of yards.

This seems, indeed to be one of the plants whose whereabouts are oftenest revealed by what we are wont to term a “happy accident” as for instance, when we are lying stretched on the ground, resting, or as we stoop, at lunch, to crack an egg on the toe of our shoe. I know of one excellent collector who spent a whole day looking for it diligently in what he thought to be a likely spot but without success when finally, just before the time for return came, as he was half crouching on the ground, scarcely thinking now of *Schizaea*, its fronds suddenly flashed upon his sight, right at his feet.

The sterile fronds of *Schizaea pusilla* are evergreen so the collector may perhaps best detect it in winter selecting days for his search when the ground is pretty clear of snow. The surrounding vegetation being at that time dead the little corkscrew-like fronds stand out more prominently. The fertile fronds die before winter sets in but their brown stalks frequently nevertheless remain standing long after.—C. F. Saunders in 55 *Linnaean Fern Bulletin*, Vol. 4.

PTERIDOGRAFIA.

A NEW FERN PEST.—According to the *British Fern Gazette* a new pest threatens the specimens of those who collect living plants. This is the larva of a small weevil which gets into the stipes of the ferns and burrowing downward into the heart of the rhizomes soon cause the death of the plant. The weevil is of Australian origin, probably introduced into Britain with imported plants. Its scientific cognomen is *Syagrius intrudens*. At first its depredations were confined to ferns under glass, but more recently it has taken to the ferns in the wild state. This, however, is not the only enemy of the ferns that British growers have to contend with. Another small beetle known as the vine weevil (*Otiorhynchus sulcatus*) is fond of the plants both in the adult and larval stages, but the newcomer has already developed a reputation for destructiveness that places it first as a fern pest.

WALKING FERN AND LIME.—Nearly everybody who cultivates the walking fern (*Camptosorus rhizophyllus*), thinks it necessary to supply it with a quantity of old mortar, quick-lime or pieces of limestone under the impression that the fern cannot live, or at least cannot thrive without a considerable amount of calcium in the soil. As a matter of fact it has been reported on sandstone, shale, gneiss and granite and may possibly grow on others. Its noticed preference for limestone is apparently not due to its dependence on calcium but rather to the fact that it is more nearly adjusted to the plant covering of limestone rocks than it is to others. It will grow in any good garden soil, but in such situations it must be protected from its enemies, the ordinary weeds of cultivation, which otherwise would soon run it out. The same thing is true of many plants besides ferns. The cactus plant that cheerfully endures the intense insolation and frequent drouth of the sand barrens, succumbs very soon to the grass and weeds when planted in rich soil.

STIPE OR STIPES.—When it comes to the designation of the stalk of a fern leaf, there is a wide difference in the way British and Americans regard it.

Americans invariably speak of a single stalk as a stipe and they may be somewhat astonished, upon referring to a dictionary, to find that while stipe is given as a legitimate word, it comes direct from the latin *Stipes* which the Britons, with perhaps a more classical education, are accustomed to use. In America the plural of stipes is stipes or, rather, the plural of stipe is stipes; but in England the plural of both stipe and stipes is *stipites*. In certain uncultivated parts of our own country the singular form of the word species is given as specie; but when we smile at some countryman's description of a specie of fern, our merriment may be somewhat tempered by the thought that we still say stipe instead of stipes. If we could only believe that we use stipe with full knowledge of its derivation, it would not seem so bad, but it is very evidently a case of plain ignorance.

APOGAMY IN PELLAEA.—Apogamy, or the production of a new sporophyte from the gametophyte without the union of egg and sperm, used to be considered a rather rare phenomenon, but as more study is given the matter, it begins to seem fairly common. Several years ago Woronin reported apogamy in *Pellaea flavens*, *P. niveus* and *P. tenera* and still more recently W. N. Steil of the University of Wisconsin reported the same condition in our native *Pellaea atropurpurea*. In Steil's specimens the young sporophytes were borne on the prothallus lobes near the notch. The same investigator is now working on apogamy in other species. A note in a recent number of this magazine asked for spores of *Pellaea gracilis* (*Cryptogramma Stelleri*) for this purpose. 57

LYCOPodium LUCIDULUM POROPHYLUM.—In the *Ohio Naturalist* for April Prof. J. H. Schaffner devotes several pages to a discussion of the specific distinctness of forms allied to *Lycopodium lucidulum* and comes to the conclusion that *Lycopodium porophyllum* is a good species. If one is to judge by appearances alone, there can be no question as to *L. lucidulum* being different from *L. porophyllum* but if the different appearances that plants put on under different conditions of warmth, light and moisture are to be considered then there are a number of fern species in this country in need of a name. Compare *Woodsia obtusa* grown on a sunny cliff with the same species grown on a moist one, or *Equisetum arvense* in woods and on railway banks. Nobody at present can say positively whether the form

called *porophyllum* is a species or not. If it can be grown in moisture and shade while still retaining its characters, or if its spores will produce plants like the parent when sown in moist shades, then the case should be considered closed. Meanwhile, if one were to imagine a dry ground form of *L. lucidulum* what kind of a plant would he construct? Perhaps prostrate stem shorter; branches in a denser tuft, shorter; leaves less notched, smaller; whole plant yellower. Well, that is the description of *L. porophyllum*!

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AFFINITIES OF TAENITIS.—The genus *Taenitis* is one that has always puzzled botanists. It was once placed in the tribe Grammitideae along with such genera as *Notholaena*, *Brainera*, *Meniscum*, *Vittaria*, *Hemionitis* and *Drymoglossum*, and it has also been considered sufficiently distinct to stand as the type of a tribe named for it, while recently it has been considered as a member of the tribe *Polypodicae*. Now comes E. B. Copeland in the *Philippine Journal of Science* and gives the genus another turn and this time places it in the Davallieae largely upon the relationship shown by the internal structure of the stem and the character of the scaly covering. It is likely that the new manipulator of the genus is as near right as anybody. The main thing is to discover what are the real indications of relationships. With some students it is venation, with others the shape and position of the indusium, with others the character of the vestiture and still others may have other rules by which to judge. When we agree upon the proper earmarks, anybody ought to be able to put the ferns in their proper groups.

SPOROPHYLL ZONES.—The fact is well known that some of the club-mosses, notably the shining club moss (*Lycopodium lucidulum*) and the fir club-moss (*L. Selago*), bear their sporangia in bands or zones that alternate with regions on the stem in which there are no sporophylls, but it does not seem to be equally well recognized that the same phenomena are found pretty generally among the ferns. If one will examine the crowns of the cinnamon fern, it will be readily seen that sporophylls and vegetative leaves form alternating circles. Curiously enough, the fertile fronds, which appear at maturity within the circle of sterile leaves, really belong to the outer circle, as befits the group that is to develop first. The sensitive and ostrich ferns are other species in which the zones of fronds are very distinct. So pronounced is this, and so far has each kind developed before unfolding,

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that each is usually incapable of taking up the functions of the other in cases where the destruction of one kind makes such exchange necessary or desirable. From efforts on the part of the plant to supply vegetative tissue to leaves designed originally for spore-bearing, only, we owe the various “obtusilobata” forms occasionally reported. The differences in zonation here mentioned are most pronounced in ferns with dimorphic fronds, but evidences of the same thing, more or less distinct may be found even in those ferns that have the fertile and sterile fronds essentially alike in outline. As a usual thing, the spore-bearing leaves are produced after the vegetative leaves have unfolded and when we find a plant in full fruit in late summer, that lacked spores in spring, it is due to the developing of the fertile leaves later. This is especially true and most noticeable in ferns that produce their fronds in crowns, but even in those species with running rootstocks, we commonly find evidences of zonation. Following out the idea of zonation we find among many of the fern allies that not only are the sporophylls assembled in zones but the zones terminate the central axis or branch. Under such circumstances the shoot begins to take on many of the characteristics of the flower and if we allow the definition of a flower as a shoot beset with sporophylls, it really is a flower. In the plants in which the flower comes to its highest development this structure is essentially a group of two kinds of sporophylls set round with sterile leaves called petals and sepals. Did ferns, instead of selaginellas, produce two kinds of sporophylls, the whole fern plant with its crown of fronds, would be very like a flower.

INDEX TO RECENT LITERATURE.

Readers are requested to call our attention to any errors in, or omissions from, this list.

CLUTE, W. N. *Nephrodium deltoideum*. illust. Fern Bulletin, Ja. 1912.

CLUTE, W. N. *Rare Forms of Fernworts*.—XXI. *Another Form of the Christmas Fern*. illust. Fern Bulletin, Ja. 1912.—*Polystichum acrostichoides* f. *lanceolatum* described and illustrated.

DARLING, N. *Observations on some Lycopodiums of Hartland Vt.* illust. American Fern Journal, Ap. 1912.

DODGE, C. K. *The Fern-flora of Michigan*. Fern Bulletin, Ja. 1912.—Fifty-eight ferns and thirty-one fern allies listed with notes.

COCKAYNE, L. *Some Noteworthy New Zealand Ferns*. illust. Plant World, Mr. 1912.

HILL, E. J. *Additions to the Fern-flora of Indiana*. Fern Bulletin, Jl. 1912.—New stations for several species.

HILL, E. J. *The Rock Relations of the Cliff Brakes*. Fern Bulletin Ja. 1912.

HOPKINS, L. S. *Lycopodium Selago from Ohio*. illust. American Fern Journal, Ap. 1912.—A form of *L. lucidulum* mistaken for the rarer species.

PRESCOTT, A. *The Osmundas*. Fern Bulletin, Ja. 1912.

SAFFORD, W. E. *Notes of a Naturalist Afloat*.—III: illust. American Fern Journal, Ap. 1912.—Occasional mention of common Ferns.

SCHAFFNER, J. H. *The North American Lycopods without Terminal Cones*. illust. Ohio Naturalist, Ap. 1912.—*Lycopodium porophyllum* regarded

as of specific rank.

WINSLOW, E. J. *Some Hybrid Ferns in Connecticut*. American Fern Journal,
Ap. 1912.

EDITORIAL.

The last number of this magazine—that for October 1912—will be a comprehensive index of the publication for the past ten years. This, with the index to the first ten volumes, will form an exceedingly valuable index to the fern literature of America, covering, as it does, the whole period of popular fern study. It begins some years before the appearance of any popular fern book and has either published entire all important articles issued since or given a summary of them. Mr. S. Fred Prince, long a member of the Fern Society is already at work on the index and we hope to issue it not later than the end of the year.

* * * * *

Further information received from the purchaser of the complete set of this magazine recently sent to Germany, apprises us of the fact that the set is not to remain in Europe. It was purchased for a customer in South America (Argentina), therefore the set owned by M. C. Belhatte at Paris is the only one in Europe. The recent set is also the only complete set in South America, and there are not, so far as we are aware, complete sets in other parts of the Old World though the set at the Tokyo Botanical Garden ought to be nearly complete and the set owned by D. Leroy Topping at Manila lacks only two numbers.

* * * * *

Next year it will be too late to get odd numbers to complete files that lack them. When this magazine goes out of business we shall retain only complete volumes. This is the time for all who need odd numbers to ask for them. We have recently advertised to send any back volume later than volume 9 for 50 cents and this offer will hold good until the end of the year. After that time, single volumes cannot be had unless we happen to have a surplus. We are willing to replace soiled, torn or missing numbers free if requested to do so at once, and the fact that odd volumes

will soon be unobtainable should incline all whose sets are incomplete to add the missing volumes while they can.

BOOK NOTES.

In anticipation of the consolidation of this magazine with *The American Botanist* at the end of the year, some very extensive improvements in the new magazine are to be made. Among the more important are a better grade of paper, the use of numerous illustrations, and the addition of enough pages to make it the largest magazine for the price in America. With the beginning of 1913 a department of ornamental gardening will be included in which the cultivation of our showy wild-flowers will receive adequate treatment. This magazine will continue the matter relative to ferns now appearing in *The Fern Bulletin* and all manuscripts used will be paid for. No reader of *Fern Bulletin* should fail to subscribe for the new *American Botanist* if they wish to keep abreast of the times in botany. Those who subscribe for 1913 before November 20th, will receive the November issue free.

Messrs. Ginn & Co. have nearly ready for publication a book on Agronomy by the editor of *The Fern Bulletin* which should be of interest to all who have anything to do with cultivating plants. Although the book is intended as a school book to be used in connection with gardening courses, the fact that it not only gives directions for planting and cultivating kitchen vegetables and flowering plants, but explains the principles upon 63 which such directions hinge, will make it of much value to the gardener whether amateur or professional. The book, however, is not a mere gardening manual. It discusses soils and their origin, the fundamentals of landscape work and plant breeding, and the effects of heat, light and moisture upon plants in general. There will also be more than 200 illustrations.

FERNS WEIGHING A TON.—In the tropics ferns often attain the height of small trees, but their trunks are usually so slender that they never are of any great weight. For the heaviest trunks we must look among lowlier species, where the circumference of the short trunk in some cases is so great that immense

weights are attained. In Australia and New Zealand there grows a relative of the common cinnamon fern named *Todaëa barbata* which quite takes the palm in this respect. The trunks are great rounded mosses five or six feet high and at least twenty feet in circumference, most of the upper surface being beset with living fronds. Specimens have been found with trunks that were estimated to weigh more than a ton and a half.

All the American Fern Books

Ferns of Kentucky 12mo.	(out of print)
Ferns of North America 2 vols. 4to.	(out of print)
Fern Collector's Handbook 4to.	(out of print)
Ferns in their Homes and Ours	(out of print)
Our Ferns in Their Haunts 8vo.	\$2.15
How to Know Ferns 12mo.	1.60
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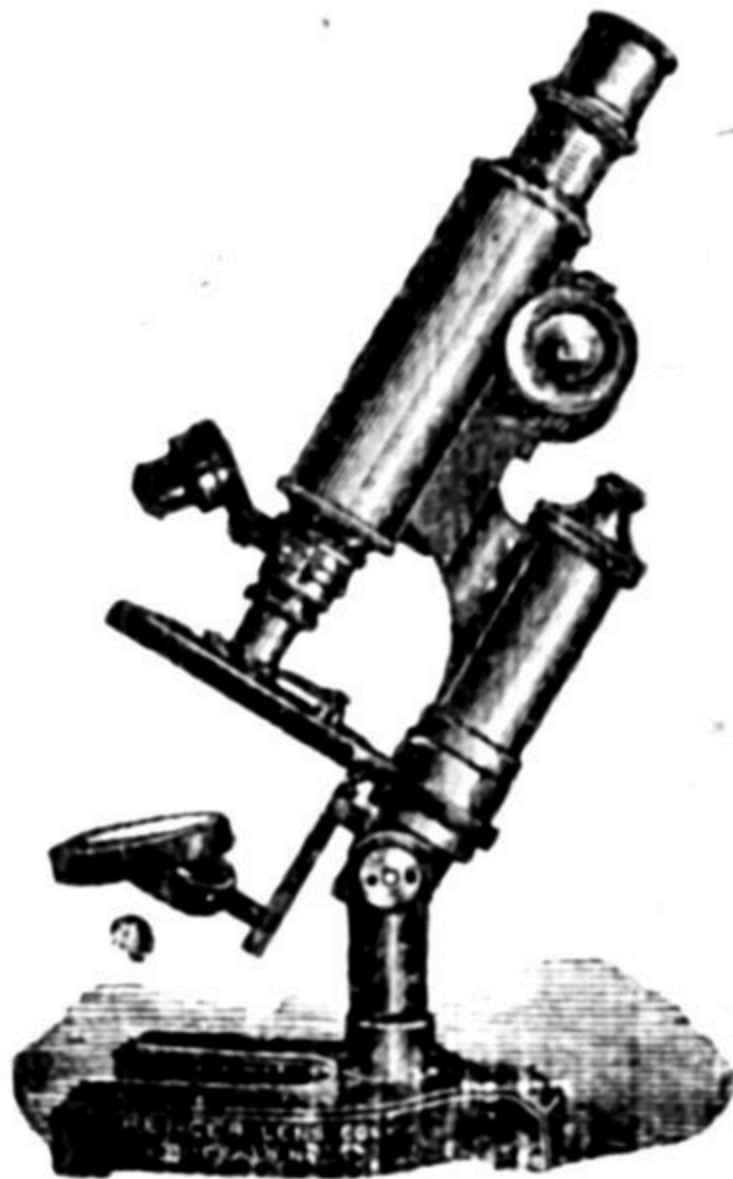
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ANOTHER COPY

We have obtained another copy of Williamson's "Ferns of Kentucky" which we offer for \$4.50 postpaid. This volume, the first American fern book, compares very favorably with the most recent. It contains fifty-nine full page plates reproduced from etchings, and is the only fern book so illustrated. More than 150 pages of text give an account of the species. Valued highly for its unique position among fern books, and hard to get at any price, this copy will not remain on hand long. Order at once.

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BULLETIN, APRIL 1912 ***

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